CCTE R82 Lab Scripts: lab menu and improvements

Handout for instructors and students · scripts version 2.2.2 · 8 November 2026 · Don Paterson

Each CCTE lab starts with a **lab script** that creates the problem you troubleshoot. In this class the scripts have been updated and a menu, **labmenu**, has been added. **Nothing in the lab guide changes:** every command in the book still works exactly as written and sets up the same lab. The menu is simply an easier way to run the same commands.

1. Two ways to run a lab script

	As in the lab guide	With the menu
Where	A-SMS (A-GW-01/02 for Lab 4A), Expert mode	Same
How	`cd /home/admin/CCTE_Scripts` `./labstart lab8a`	`labmenu`, choose the lab, then **1) Start**
Stop / Revert	`./labstop lab8a` `./labrevert lab8a`	**2) Stop** / **3) Revert**
Verbose	`./labstart lab8a -v`	press **v** (Verbose ON in the header)
Where a lab stands	not available	**4) Status** (read-only check)

The menu prints the exact command it runs, for example `Running: ./labstart lab8a`, so what you see always matches the book. After Start, Stop or Revert the menu waits for the policy installations and shows each result with the reason for any failure, the same as Recent Tasks in SmartConsole. It also waits for running installations before it starts a new action.

2. Using labmenu

1. Log in to A-SMS as admin (from A-GUI with PuTTY or SSH). 2. Type `expert` and the admin password. 3. Type `labmenu` and press Enter. 4. Choose a lab number, then an action. 5. b goes back, q quits.

```
CCTE R82 Lab Scripts   host: A-SMS   scripts: 2.2.2
Verbose: off    folder: /home/admin/CCTE_Scripts
Labs on this host:
   1) LAB2A  Troubleshoot Security Management Server Processes  [not running]
   2) LAB2B  Troubleshoot Management High Availability
   3) LAB3A  Troubleshoot SmartLog  [not running]
   4) LAB5A  Debug the Security Gateway Kernel
   5) LAB6A  Debug Unified Policy Inspection
   6) LAB8A  Debug Identity Awareness
   7) LAB9A  Troubleshoot Site-to-Site VPN
  v) Verbose on/off   h) Help   q) Quit
```

Lab 4A is on the gateways: run `labmenu` on A-GW-01. Lab 2B uses both management servers: start it on A-SMS first, then on A-SMS-02, as the lab guide says.

3. What the messages mean

You see	Meaning	What to do
`Setup completed. Kindly check policy installation status in SmartConsole`	The script did its job.	Carry on with the lab. In some labs (for example 8A) a policy install is **meant** to fail: that is the problem to troubleshoot.
`Lab lab8a is already set up (...). Nothing was changed.`	**New.** The lab was set up by an earlier run.	Carry on with the lab. To start it again from the beginning, use Revert, then Start.
`ERROR: ...` then `Setup ended`	Something failed. Nothing was changed.	Read the reason on the next line. Close SmartConsole or publish your changes, then try again, or ask the instructor.
`Note: lab3a has no revert script...`	**New.** Only labs 2B and 8A have a revert script; for the others Revert runs the stop script.	Run Start again to set the lab up afresh.
`labXa is already running. Do you want to restart it?`	Labs 2A, 3A and 4A run in the background until stopped.	Answer n to leave it running.

You see	Meaning	What to do
`Interrupted - discarding changes and logging out`	**New.** You pressed Ctrl-C.	Nothing is left locked. Start the lab again.

4. What is different from the original Check Point scripts

Same commands, same lab problems, same normal messages. The changes only show when something unusual happens:

Situation	Original scripts	Updated scripts (2.2.2)
Running `labstart` a second time (6A, 8A, 9A)	Prints only `Setup ended`, with no reason. 9A shows an error.	Explains that the lab is already set up and how to start it again
`labrevert` on a lab without a revert script	Silently runs the stop script	Says so, and tells you to run Start again
Ctrl-C during a lab script	Can leave objects locked in SmartConsole, or (Lab 2B) A-SMS-02 cut off	Discards the session and logs out; Lab 2B restores the route to A-SMS-02
A failure	Some failures print no reason, or report success	Always `ERROR:` plus the reason from the management server
`-v` before the lab name	Only the launcher is verbose	Works before or after the lab name
Mistyped lab name	Unclear error	Clear error with examples
Running `labstop lab9a` twice	Error on the second run	Completes normally
Revert then Start straight away	Installs can collide and fail	The menu waits for the first installs to finish

5. Lab notes: where the lab and the book can differ

Lab / step	What you may see	What to do
2B, step 2B-1.3	Page 160 shows A-SMS as **No Communication** instead of a conflict.	Go back to the **first** A-SMS window, the one showing `Type "exit" to restore connectivity`, and type `exit` there. Do not open a new session. Allow 3 to 5 minutes for the conflict to show.
2B, step 2B-2.32	View Sessions also lists an (Unnamed) **WEB_API** / Management API session from 127.0.0.1 with 0 locks and 0 changes.	A background system login, not another administrator. Ignore it or disconnect it; it closes by itself after about 10 minutes.
2B, step 2B-2.37	**Sync Peer** is greyed out; A-SMS shows Standby, *Successfully synced at ...*	The automatic sync has already run. Skip 2B-2.37 and 2B-2.37a and continue.
8A, Task 8A-1	`Setup ended` with no error (original scripts) or *already set up* (2.2.x).	Lab 8A was already started. Carry on from Task 8A-2, or Revert then Start to begin again.
8A, Task 8A-2	Bravo-Standard install fails on B-GW: *rule with Access Role includes targets that do not support Access Role*.	Intended. This is the problem Lab 8A asks you to troubleshoot.
Any lab	A policy install fails on A-GW-Cluster straight after a Revert.	The next action started before the previous installs had finished. Use the menu, which waits, or wait a minute between actions.

6. For instructors

Install on A-GUI in PowerShell (about 20 seconds per host; asks for the lab admin password once and does not store it):

```
irm https://raw.githubusercontent.com/Caminoltd/CCTE-R82-LabScripts/main/Install-CCTEScripts.ps1 | iex
```

It updates A-SMS, A-SMS-02, A-GW-01 and A-GW-02, keeps the original scripts in `/home/admin/CCTE_Scripts/.orig` and prints one OK/FAIL line per host. **Undo:** run the same command with `-Restore` (see the README on GitHub). **Check a host:** `cat /home/admin/CCTE_Scripts/VERSION` shows

`2.2.2`.

Good to know. Admin logs in to clish on the appliances, so students must type `expert` first. Verbose output and the last API reply are in `/tmp/labXa.json` and `.err` (menu option 5). If Lab 8A start reports a timeout querying `alpha.cp`, AD is still waking up after a lab resume: wait a minute and run it again. The ATC portal download of `lab8a_start` has a typo (TikTokd, line 74); the lab image and this set are correct.

Source code, change list and originals: github.com/Caminoltd/CCTE-R82-LabScripts.